Akoizo Original Practice - Form 3

Division C | Botany | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) The Casparian strip particularly restricts…

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

A. Unregulated apoplastic entry

B. All photosynthesis

C. Every phloem tube

D. All root growth

Answer: _____

2. (2 points) Ferns reproduce through…

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

A. Spores with a gametophyte stage

B. Flowers and fruit universally

C. Only seeds

D. No meiosis

Answer: _____

3. (2 points) A fungal structure on a lesion is an example of…

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

A. A sign

B. Only a symptom

C. A root system

D. A light-response plateau

Answer: _____

4. (2 points) Do all plant cells have many chloroplasts?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

A. Only every dead xylem cell

B. No

C. Yes

D. Only every root cell

Answer: _____

5. (2 points) Pollination and fertilization are…

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

A. Both meiosis

B. Different events

C. Synonyms

D. Both always seed dispersal

Answer: _____

6. (2 points) Oldest known fossil equals exact origin time?

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

A. Only for pollen

B. Not necessarily

C. Always

D. Only if the fossil is large

Answer: _____

7. (2 points) A herbarium primarily preserves…

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

A. Every soil variable

B. A full evolutionary tree automatically

C. Specimens and collection evidence

D. A continuous gas-exchange record

Answer: _____

8. (2 points) Hormone effects depend on…

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

A. Only leaf color

B. No concentration effects

C. Tissue and context

D. Only the hormone name

Answer: _____

9. (2 points) A high-light plateau proves one unique cause?

Unit: Photosynthesis, carbon fixation, and plant biochemistry.
Worked-case data: A leaf fixes 12 relative carbon units and releases 4 through the specified respiratory balance. What is net gain?

A. Yes, always no chlorophyll

B. Yes, always no roots

C. No

D. Yes, always light shortage

Answer: _____

10. (2 points) Vegetative propagation commonly helps preserve…

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

A. No traits

B. Every environmental effect

C. A selected genotype

D. Maximum new genetic variation necessarily

Answer: _____

11. (4 points) Compare apoplast and symplast.

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

Answer: _____
_____
_____

12. (4 points) Name two major seed-plant innovations.

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

Answer: _____
_____
_____

13. (4 points) Distinguish symptom and sign.

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

Answer: _____
_____
_____

14. (4 points) What is an apical meristem’s role?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

Answer: _____
_____
_____

15. (4 points) What commonly forms triploid endosperm?

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

Answer: _____
_____
_____

16. (4 points) Name a terrestrial tradeoff.

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

Answer: _____
_____
_____

17. (4 points) Name two plant ecosystem functions.

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

Answer: _____
_____
_____

18. (4 points) What does a directional-light control test?

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

Answer: _____
_____
_____